Data-Driven Prediction of AI Optimization Response

Scaling-state descriptors, future response magnitude and useful-progress targets

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Abstract

This paper develops a prediction procedure for future cross-scale optimization response using an observed n/s scaling-state representation. It distinguishes moment-order extrapolation from temporal forecasting, future response magnitude from future loss improvement, and observed-policy prediction from counterfactual action response. Linearized descriptor fitting, response uncertainty and target-specific WPM landscapes provide a mathematical workflow. Exact examples show why the norm of a predicted mean is not the expected response magnitude and why a learned structural predictor is not fed the optimizer's true governing equation.

Research lineage and mathematical status. This study develops a WPM mathematical extension with reference to the author's published WPM architecture [1] and foundational n/s scaling-spectrum framework [2]. The n/s paper concerns conditional moment-spectrum parameterization; the WPM paper provides the broader architectural reference. The results proved here are mathematical consequences of stated assumptions, usually applications of standard linear algebra, probability or optimization. The WPM contribution is their organization into an explicit workflow. Modeling constructions and applicability conditions are identified separately; no universal forecasting superiority is asserted.

Predictor inputs. The forecasting layer described here is data driven: it learns from observed histories and matured outcomes. It is not fed the system's true governing equation or unobserved future states at issuance. Physical equations used below define observables, explain a limiting example or specify a separately labeled comparator. Optional supplied-map inverse problems and physics-informed variants must be declared separately; they are not evidence that the default forecast knows the generator.

1 The response query and the observation boundary

Let $\mathcal{F}_t$ contain pre-response training observations and known context. A future perturbation or optimization block can change the measured structural state. The query is to predict a declared response from information available before that block. The future event-period measurements define the target and must not enter the pre-event features.

A controlled future action may be known at issuance. If actions vary, include the action in the query or define the policy over which prediction is averaged. A response learned under one

fixed perturbation is not automatically a prediction under every learning-rate, batch-size or data change.

Write the pre-event structural descriptor as Θ_t and the later descriptor measured under a fixed protocol as $\Theta^+_{t,P}$. A magnitude target is

$$R_{t,P} = \left\| D^{-1}(\Theta^+_{t,P} - \Theta_t) \right\|, \tag{1.1}$$

where D supplies declared units/scales. This is a quantitative response target, distinct from a binary transition label, future validation-loss improvement and final model quality.

2 An actually fitted n/s representation

For admissible moment orders $q \neq 0$, the released family is

$$H(q,t) = \frac{|q|^{n_t - 1}}{s_{0,t} e^{k_t q}}, \qquad s_{0,t} > 0. \tag{2.1}$$

With valid positive spectrum estimates, logarithms give

$$\log H(q,t) = (n_t - 1)\log|q| - \log s_{0,t} - k_t q. \tag{2.2}$$

Fit the design columns $(\log|q|, 1, q)$ using a suitable observation-error model. Descriptor coordinates can be $(n, \log s_0, k)$, but if an archived target used (n, s_0, k) that is a different metric geometry. A response norm changes under a nonlinear coordinate transform unless the target map is adjusted explicitly.

Proposition 2.1 (Descriptor identifiability within the stated family)**.** *Noiseless coefficients of the log-spectrum model are unique exactly when its design has full column rank.*

Proof. Two coefficient vectors give the same observations precisely when their difference lies in the design nullspace. Full column rank makes that difference zero. Convert the intercept b into $s_0 = e^{-b}$ and the q coefficient c into $k = -c$. □

Near-zero fluctuations can invalidate negative moments; heavy tails can destabilize high positive moments. The measurement method, admissible order range and fitting uncertainty must accompany the descriptor. A compact loss-slope/scale proxy is not an actual n/s spectrum merely because it serves a similar predictive role.

3 Moment-order prediction is not time prediction

The native sequential rule fits the spectrum at available lower moment orders, predicts a new order, then reveals it and checks the prediction. For example, fitting on $|q| \leq 2$ and predicting $H(\pm 3)$ is extrapolation across q. It does not provide a differential equation for $n(t), s_0(t), k(t)$.

Temporal response forecasting instead fits a relation such as

$$\widehat{R}_{t,P} = g_\beta(\Theta_t, E_t, C_t, P), \tag{3.1}$$

where E_t is a declared spectrum consistency/error summary and g_β is trained on earlier complete response examples. The learned regression is the prediction model. It receives no true future optimizer equation or hidden response label at issuance.

Worked example 3.1 (Same spectrum state, different future action)**.** Two runs may have the same current descriptor but receive different future perturbations. If the action affects response, a predictor omitting it estimates a mixture over the action policy, not the response to each action. Identical current n/s coordinates alone do not force identical future evolution.

4 Response magnitude and vector uncertainty

One can model the descriptor displacement vector Δ and then transform its joint distribution to R, or fit R directly. The two methods need not give the same fitted result.

Theorem 4.1 (Mean response cannot be obtained by taking only a mean vector norm)**.** *For integrable random vector* Δ*,* $\|\mathbb{E}\Delta\| \le \mathbb{E}\,\|\Delta\|$*. If second moments exist,*

$$\mathbb{E}\,\|\Delta\|^2 = \|\mathbb{E}\Delta\|^2 + \operatorname{tr}\operatorname{Cov}(\Delta). \tag{4.1}$$

Proof. The inequality is Jensen's inequality for the convex norm. For the identity, expand $\Delta = \mu + (\Delta - \mu)$, take squared norms and expectations, and use zero mean of the centered displacement. The expected centered squared norm is the trace covariance. □

Worked example 4.2 (A missed response from vector averaging)**.** Suppose $\Delta = (1, 0)$ or $(-1, 0)$ with equal probability. Its mean is zero, but response magnitude is always one. A mean-vector forecast followed by a norm gives zero and misses the response. A joint predictive model or direct magnitude model preserves it.

For a metric M, replace squared norm by $\Delta^\top M\Delta$; expected value is $\mu^\top M\mu + \operatorname{tr}(M\Sigma)$. The scaling convention is therefore part of the prediction target, not merely a plotting choice.

5 Descriptor measurement error and target noise

If estimated pre/post states are $\widehat{\Theta} = \Theta + e$ and $\widehat{\Theta}^+ = \Theta^+ + e^+$, their displacement error is $e^+ - e$. Overlapping measurement windows can correlate those errors. A large measured response can partly reflect fitting noise, and a small raw norm can conceal uncertainty.

State whether the target is the observed descriptor response or a latent noise-corrected response. Propagating the fitted parameter covariance through a norm is nonlinear near zero; a first-order derivative can be unstable there. Joint samples, an appropriate latent measurement model or a directly defined observable target may be more suitable than treating all fitted descriptors as exact.

6 Noise-adjusted response and noisy predictive features

Suppose the measured displacement is $\widehat{\Delta} = \Delta + e$, with conditional zero-mean error and known conditional covariance Σ_e. Then

$$\mathbb{E}[\left\|\widehat{\Delta}\right\|^2 \mid \Delta] = \|\Delta\|^2 + \operatorname{tr}\Sigma_e. \tag{6.1}$$

Expand the square to prove the identity. Subtracting $\operatorname{tr}\Sigma_e$ gives an unbiased estimator of squared response under these assumptions, but it can be negative on a particular observation. Clipping at zero changes that unbiasedness. Taking its square root is not an unbiased estimator of the response norm. This distinction matters most near a zero-response boundary.

The pre-response descriptor is also a noisy regressor. In the scalar model $Y = \beta X + \epsilon$ with observed $\widetilde{X} = X + U$, independent zero-mean U and ϵ, fitting ordinary least squares through centered variables has population slope

$$\frac{\operatorname{Cov}(\widetilde{X}, Y)}{\operatorname{Var}(\widetilde{X})} = \beta \frac{\operatorname{Var} X}{\operatorname{Var} X + \operatorname{Var} U}. \tag{6.2}$$

The covariance numerator loses U by independence, while the denominator adds its variance. Hence plugging uncertain descriptors into a regression as exact can attenuate coefficients. A measurement-error or latent-state model is a separate modeling choice, not a correction supplied by pocket labels.

7 A nonnegative predictive response distribution

Response magnitude R is nonnegative. A direct distributional predictor should respect that support, for example with a nonnegative law, an explicit atom at zero plus a positive continuous component, or a declared transformation. A Gaussian residual model for a positive target can assign negative probability mass and needs interpretation or a different construction.

If the fitted object is a CDF F on $[0, \infty)$, $\mathbb{P}(a \le R \le b) = F(b) - F(a^-)$ for $0 \le a \le b$. Mixtures of compatible response CDFs remain valid CDFs under nonnegative normalized weights. Their validity as distributions is not a calibration guarantee.

Proposition 7.1 (Intersecting an uncertainty set with known support)**.** *If $\mathbb{P}(R \in A) = 1$, any random set C has $\mathbb{P}(R \in C \cap A) = \mathbb{P}(R \in C)$. Thus intersecting a valid response interval with $[0, \infty)$ preserves its actual coverage.*

Proof. The two membership events differ only on $\{R \notin A\}$, a null event. Their probabilities therefore agree. □

This does not make an invalid interval valid. If support, observation units or the response definition are wrong, the premise fails. The statement also does not assert conditional coverage for every optimizer or horizon; those require their own assumptions.

8 WPM landscapes for response magnitude

For each independent W,m lens and fixed P, fit an allowed response model on historical complete examples. Store issued predictions and calculate matured squared or absolute response error. The target-specific landscape is

$$\widehat{L}_t(W, m; R, P) = \frac{\sum_{s \in \mathcal{A}_t} w_s \ell(R_{s,P}, \widehat{R}_{s,P,W,m})}{\sum_{s \in \mathcal{A}_t} w_s}. \tag{8.1}$$

The descriptor measurement convention for the response label must remain fixed while candidate forecast lenses vary. Otherwise one is optimizing across different response targets. Retain signed errors and individual pocket flow, position and existence.

Fixed is a legitimate reference. Multi combines same-response alternatives; Auto updates a qualified policy from causal information. Neither a low fitting residual nor a stable pocket certifies transfer to a new optimizer or perturbation. Empty matured history triggers a declared cold-start rule or abstention.

9 Telemetry as a comparator and optional input

Conventional pre-event telemetry may include gradient means, dispersions, loss slopes and update statistics. It is a data-driven comparator, not an equation-fed optimizer simulator. A standalone n/s predictor, a telemetry-only predictor and a combined predictor test different representations under the same target and cutoff.

Scoped evidence statement. An archived initial response comparison reported standalone n/s and conventional marginal telemetry with close errors; a subsequent independent comparison favored n/s for that particular future cross-scale response, with the combined representation varying by dataset [5]. These are learned response regressions, not proof that the spectrum equation itself evolves training in time. They do not establish a general advantage for final accuracy, generic loss recovery or every optimization target.

This supports a precise interpretation of "close to telemetry": closeness occurred in a specified initial comparison, while other comparisons differed. It is not statistical equivalence or a universal performance ordering. Telemetry remains an important matched-information reference even when it is not an input to the deployed structural predictor.

10 Useful progress is a different target

For future validation loss, define improvement $I_{t,P} = L_t^{\mathrm{val}} - L_{t+P}^{\mathrm{val}}$. Its expectation, a chance $\mathbb{P}(I_{t,P} > \delta \mid \mathcal{F}_t)$, or gain per unit compute are possible separate targets. A large structural response need not mean positive improvement; it can signal either reorganization or deterioration.

If utility is $U = I - c \cdot \mathrm{cost}$, predicting U requires the loss and cost model, not merely R. Time remaining until useful progress stops also requires a path model or a first-arrival definition. These extensions are mathematically well-defined but must not inherit the empirical support of the cross-scale response target by a change of label.

11 Action response and policy dependence

If action a is assigned under a controlled or otherwise identified design, one may model $R(a)$ conditionally. In observational training logs, the action may depend on unmeasured state; then $\mathbb{E}[R \mid A = a, X]$ need not equal the causal response under intervention. A monitor that triggers a change also alters the policy that generated its later data.

Known future actions can enter the issuance context. Unannounced future actions cannot. Counterfactual optimization recommendations require assumptions or design beyond a predictive association. The WPM resolution policy organizes the observation history; it does not remove action confounding.

12 Application procedure and solved questions

1. Fix the future response window, action policy and descriptor coordinates/metric.
2. Calculate an admissible n/s spectrum if that branch is claimed; retain consistency and measurement uncertainty.

3. Train a temporal response model from permitted pre-event features and complete future response labels.
4. Compare standalone structural, telemetry and combined representations at matched cutoffs; preserve target-specific W,m landscapes and pockets.
5. Return response magnitude or its predictive distribution with applicability, horizon and policy assumptions.
6. Keep future loss improvement, final quality and causal intervention response as separately specified problems.

Exercise 12.1. Does accurate prediction of $H(q+1,t)$ prove accurate prediction of $H(q,t+P)$?

Solution. No. The first is a moment-order extrapolation and the second a temporal forecast. A model relating future states is required for the latter.

Exercise 12.2. If displacement has mean zero and covariance $\mathrm{diag}(1,2)$, what is its expected squared Euclidean response?

Solution. It is $0+\operatorname{tr}\Sigma=3$. Mean vector zero does not imply zero magnitude response.

Scope of the mathematical exposition

Unless explicitly identified as a sourced evidence statement, numerical values are constructed arithmetic examples. Application requires specifying the observation process, fitting unknown quantities, and checking reliability for the intended population and horizon. This study shares the AI prediction collection DOI 10.5281/zenodo.23784369.

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